DATA ANALYSIS INTERNSHIP

Task 12: Final Data Analysis Project

New York City Airbnb Market Analysis

Where, What and at Which Price?
Listing Types, Areas, Prices, Demand and Hosts
of 48,895 Listings (2019)

Data Understanding • Cleaning • Exploratory Analysis • Price Model • Power BI Dashboard • Insights and Recommendations

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8 October 2026

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1 Project overview

Context. Airbnb has become a large part of the accommodation offer of New York City. Three audiences need to understand this market: **hosts**, who want to know where and at which price to list; **Airbnb**, which follows supply and demand in each area; and **the city**, which follows how short-term rentals spread across neighbourhoods.

Objective. This final project carries a complete analysis, from the raw file to an interactive dashboard, on **one dataset only, the one supplied**: 48,895 Airbnb listings of New York City in 2019. It answers the seven questions of the brief:

Q1. Which listing types are the most common?
Q2. Which areas have the most activity?
Q3. How do prices differ across areas?
Q4. Is there a relationship between price, ratings and reviews?
Q5. Which factors seem to affect price?
Q6. Which areas or listings perform best?
Q7. Are there noticeable patterns or trends?

Deliverables. The notebook `NYC-Airbnb-Analytics.ipynb` reproduces every number and figure of this report; the cleaned dataset and the cleaning log are in `data/processed/`; the result tables in `results/`; the Power BI dashboard, with its DAX measures, Power Query code and theme, in `powerbi/`; the final presentation (20 slides) in `presentation/`.

Summary of the results.

- After cleaning, **48,645 listings** of **37,283 hosts** remain (99.49% of the file). A typical night costs **$105** (median); the listings received **1.14 million reviews**.
- **Entire homes (52%) and private rooms (46%)** make the market, at a median of $160 and $70.
- **Manhattan and Brooklyn hold 85% of the listings**; Brooklyn receives the most reviews.
- **Prices fall with the distance from Manhattan**: $149 in Manhattan, $65 in the Bronx.
- **Room type and location set the price**: an entire home costs +119% and a Manhattan listing +37%, other things equal; the model explains 52% of the variation.
- **Price and reviews are almost unrelated** ($\rho = -0.05$); the file has no ratings.
- **The busiest listings are cheap rooms near the airports in Queens.**
- **40% of the listings look inactive**, and **0.9% of the hosts run 10% of the listings**.

2 Dataset information

2.1 Source and scope

The file `AB_NYC_2019.csv`, supplied for the project and the only dataset used, describes **48,895 listings** offered in New York City, one row per listing, with **16 columns**. The last review date of the file is 8 July 2019, which dates the snapshot. The listings belong to **37,457 hosts** and **221 neighbourhoods** in the five boroughs. A grey map of the city (`New_York_City_.png`) is supplied with the data and is used as the background of the maps.

2.2 Columns, meaning and types

Table 1: Columns of the file, with the type read by pandas and the missing values.

Column	Meaning	Type	Missing
id	listing identifier	integer	0
name	title of the listing	text	16
host_id	host identifier	integer	0
host_name	first name of the host	text	21
neighbourhood_group	borough (Manhattan, Brooklyn, Queens, Bronx, Staten Island)	text	0
neighbourhood	neighbourhood (221)	text	0
latitude, longitude	position of the listing	decimal	0
room_type	Entire home/apt, Private room, Shared room	text	0
price	asked price per night (US$)	integer	0
minimum_nights	minimum stay required	integer	0
number_of_reviews	reviews received since the listing was created	integer	0
last_review	date of the last review	text (date)	10,052
reviews_per_month	average reviews per month	decimal	10,052
calculated_host_listings_count	listings of the same host in the file	integer	0
availability_365	days available for booking in the next 365 days	integer	0

2.3 What the data can and cannot answer

The file describes the **offer** (where, what, at which price, under which conditions) and gives a **proxy of demand**: reviews. It has **no rating column**, no booking, no revenue and no size of the property (bedrooms, capacity). Question Q4 on ratings is therefore answered with reviews, which measure demand, not satisfaction. Prices are *asked* prices per night.

Every numeric column is strongly skewed: the median price is \$106 but the mean is \$153 and the maximum \$10,000; the median minimum stay is 3 nights but the maximum 1,250. The analysis therefore uses **medians** to describe a typical listing and rank-based tests.

3 Data cleaning process

3.1 Issues found

Table 2: Data quality audit and treatment.

Issue	Rows	Treatment
`last_review` and `reviews_per_month` missing	10,052	structural: rate set to 0, flag `has_reviews`
Listing name missing	16	*Unnamed listing*
Host name missing	21	column dropped (personal data)
Spaces around listing names	254	trimmed
`last_review` stored as text	38,843	converted to a date
Price = $0	11	removed
Price above $1,000 per night (6 at $9,999 or $10,000)	239	removed from the analysis set
Minimum nights above 365 (up to 1,250)	14	capped at 365
Duplicates (on the id and on all other columns)	0	none found
Map file `.png` that is in fact a JPEG	1 file	read by its content

The missing values are not errors: the review date and the review rate are missing **exactly** for the 10,052 listings with 0 reviews, because no review date exists for them. Setting the rate to 0 is therefore exact, and the date stays empty. A listing cannot be free, so the 11 prices at $0 are errors; the 239 prices above $1,000 (0.49%) include placeholders ($9,999, $10,000) and luxury outliers that would distort averages. They stay in the raw file. Two values that look extreme are **kept** because they are plausible: 36% of the listings have 0 days of availability (calendar closed or fully booked) and one listing has 58.5 reviews per month (a hotel-like room in the Theater District). The 989 listings that share a title with another one are different listings of the same host, not duplicates.

3.2 Cleaning log and cleaned dataset

Table 3: Cleaning log (`data/processed/cleaning_log.csv`), also shown in the dashboard.

Step	Name	Action	Rows after	Median price
1	Raw file	`AB_NYC_2019.csv` as supplied	48,895	\$106
2	Duplicates	checked on the id and on all other columns: none	48,895	\$106
3	Columns	`host_name` dropped (16 columns → 15)	48,895	\$106
4	Text	254 names trimmed; 16 missing names filled	48,895	\$106
5	Types	`last_review` converted to a date	48,895	\$106
6	Missing reviews	10,052 never-reviewed listings: rate = 0	48,895	\$106
7	Price = 0	11 listings removed	48,884	\$106
8	Price > \$1,000	239 listings removed	48,645	\$105
9	Minimum nights	14 values capped at 365	48,645	\$105
10	Derived columns	bands, host size, recency, sort orders	48,645	\$105

The cleaned dataset (`airbnb_nyc_clean.csv`) has **48,645 listings (99.49%)** and **29 columns**: the 15 original columns kept and 14 derived columns used by the analysis and the dashboard: a flag *has reviews*, price band (7 bands), minimum-stay class (1-2, 3-6, 7-29, 30+ nights), availability band (0, 1-90, 91-180, 181-270, 271-365 days), host size (1, 2-5, 6+ listings), year of the last review, days since the last review, *active in the last 12 months* (reviewed at most 365 days before 8 July 2019) and sort orders. The median price moves only from \$106 to \$105: the cleaning does not change the picture of a typical listing.

4 Exploratory analysis

4.1 Key performance indicators

Table 4: KPIs of the cleaned dataset (also the KPI cards of the dashboard).

KPI	Value	Comment
Listings	48,645	in 221 neighbourhoods
Hosts	37,283	1.3 listings per host on average
Median / average price per night	\$105 / \$141	the mean is pulled up by expensive listings
Total reviews	1,136,096	demand proxy
Listings with at least one review	79.6%	
Reviewed in the last 12 months	59.8%	the clearly active share
Median availability	44 days a year	mean 112 days
Listings of multi-listing hosts	33.9%	hosts with 2 listings or more
Most common room type	Entire home/apt	51.8%

4.2 Which listing types are the most common? (Q1)

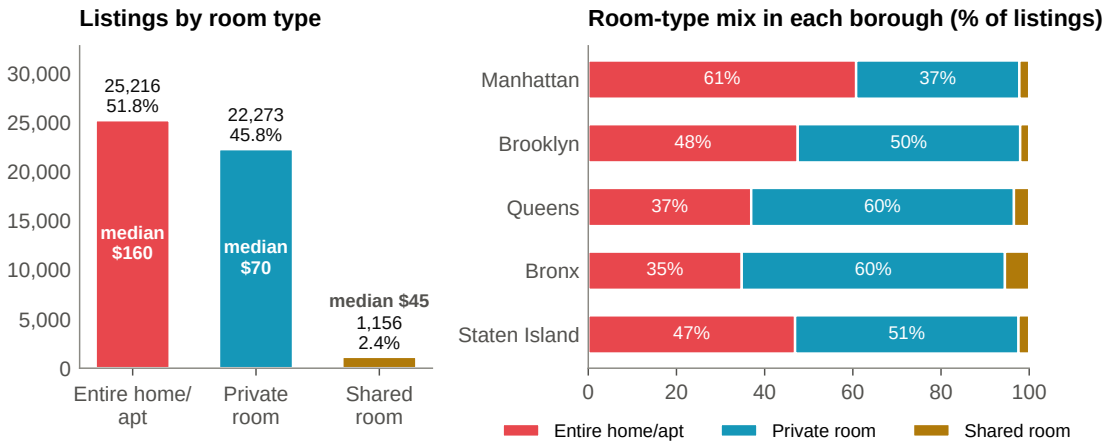


Figure 1: Listings by room type with their median price (left) and room-type mix in each borough (right).

Entire homes and apartments are the most common type (25,216 listings, 51.8%), closely followed by **private rooms (22,273, 45.8%)**; shared rooms are marginal (1,156, 2.4%). The two main types are **two different products**: an entire home costs a median $160 per night, a private room $70, a shared room $45. The mix depends on the borough: entire homes dominate in Manhattan (61%), private rooms in Queens and the Bronx (60%), where hosts rent a room in their own home. Private rooms receive almost as many reviews as entire homes (47% against 51% of all reviews) with fewer listings.

4.3 Which areas have the most activity? (Q2)

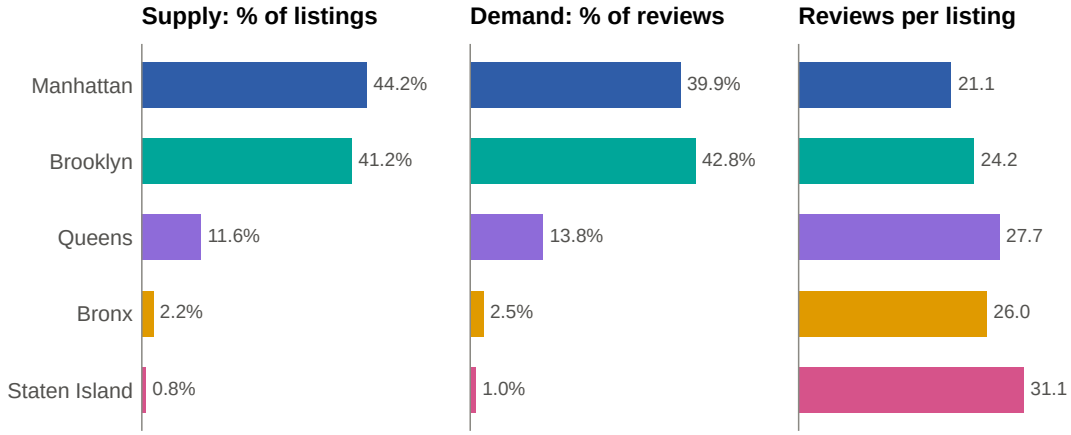


Figure 2: Supply (share of listings), demand (share of reviews) and reviews per listing by borough.

Manhattan and Brooklyn concentrate the activity: 85% of the listings and 83% of the reviews. Manhattan has the most listings (44.2%), but **Brooklyn receives the most reviews**

(42.8% against 39.9%). The small boroughs are used more intensively: **Staten Island (31.1 reviews per listing), Queens (27.7) and the Bronx (26.0)** beat Manhattan (21.1), and their median review rate (0.76 to 1.00 per month) is about three times that of Manhattan (0.29).

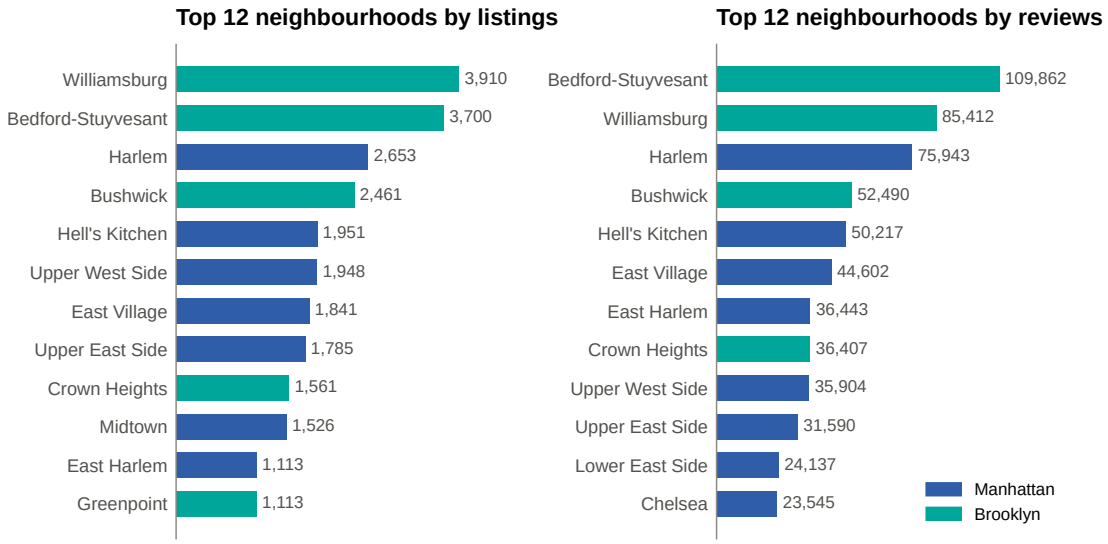


Figure 3: The 12 neighbourhoods with the most listings (left) and the most reviews (right).

The ten largest neighbourhoods hold 48% of the listings. **Williamsburg** (3,910 listings) and **Bedford-Stuyvesant** (3,700) lead the supply, and **Bedford-Stuyvesant is the most reviewed neighbourhood** of the city (109,862 reviews, ahead of Williamsburg with 85,412).

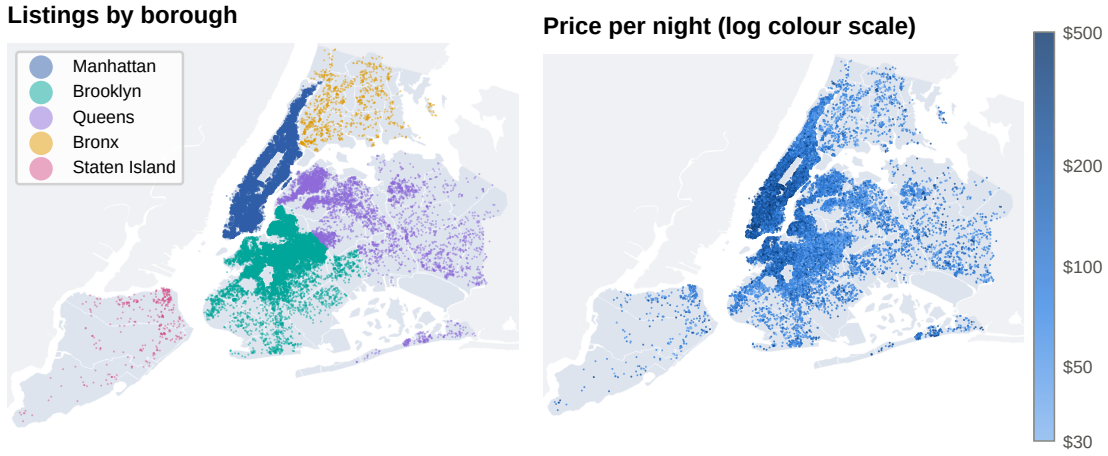


Figure 4: Map of the listings by borough (left) and by price per night (right, log colour scale), on the map supplied with the data.

The maps show the listings spreading from Midtown and Lower Manhattan into north Brooklyn and western Queens, and the highest prices in Lower and Midtown Manhattan.

4.4 How do prices differ across areas? (Q3)

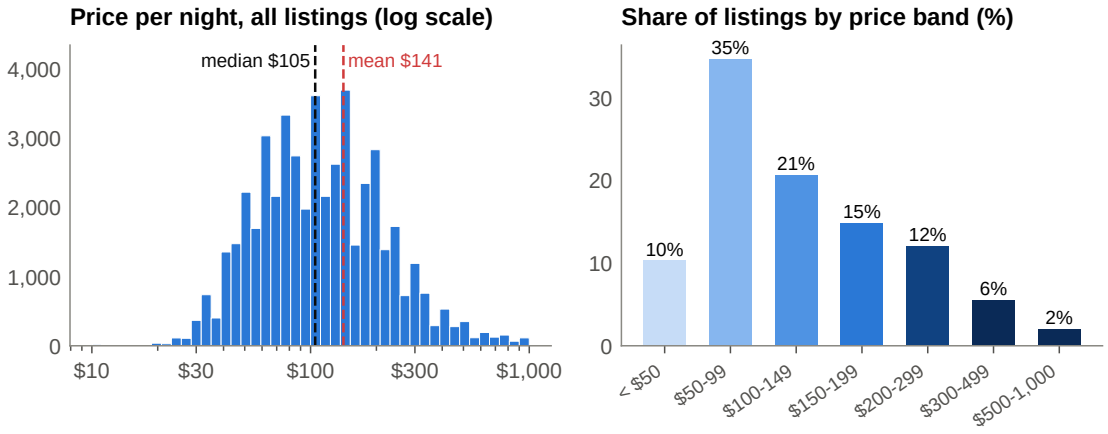


Figure 5: Distribution of the price per night (left, log scale) and share of listings by price band (right).

The median night costs **$105** and the mean $141; half of the listings ask between $69 and $175, 35% ask $50 to $99 and only 8% ask $300 or more.

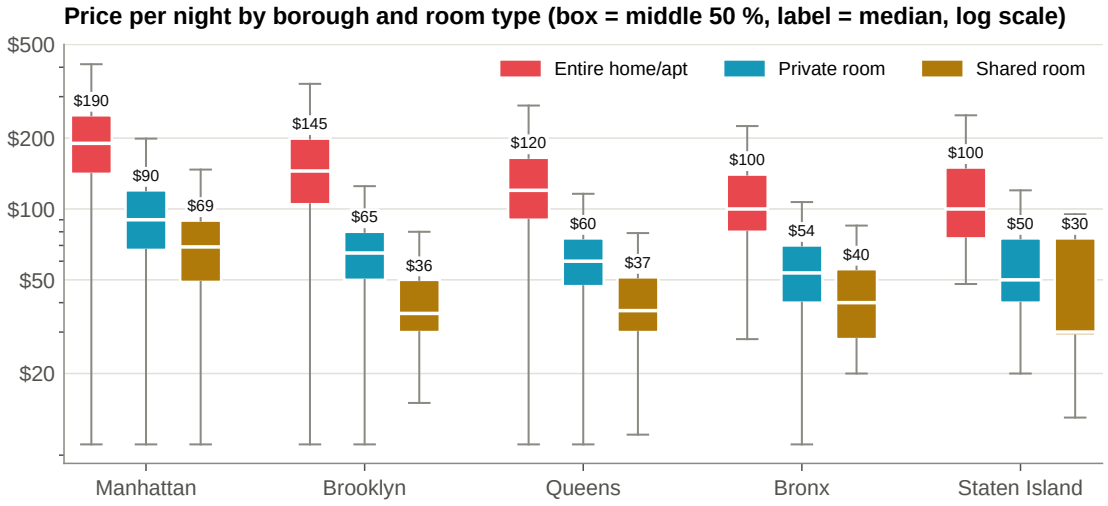


Figure 6: Price per night by borough and room type (box = middle 50% of the listings, label = median, log scale).

By borough, the median price is **$149 in Manhattan**, $90 in Brooklyn, $75 in Queens and Staten Island and **$65 in the Bronx**. For an entire home: Manhattan $190, Brooklyn $145, Queens $120, Bronx and Staten Island $100. Inside every borough an entire home costs about twice a private room, and the boxes overlap: location explains only part of the price.

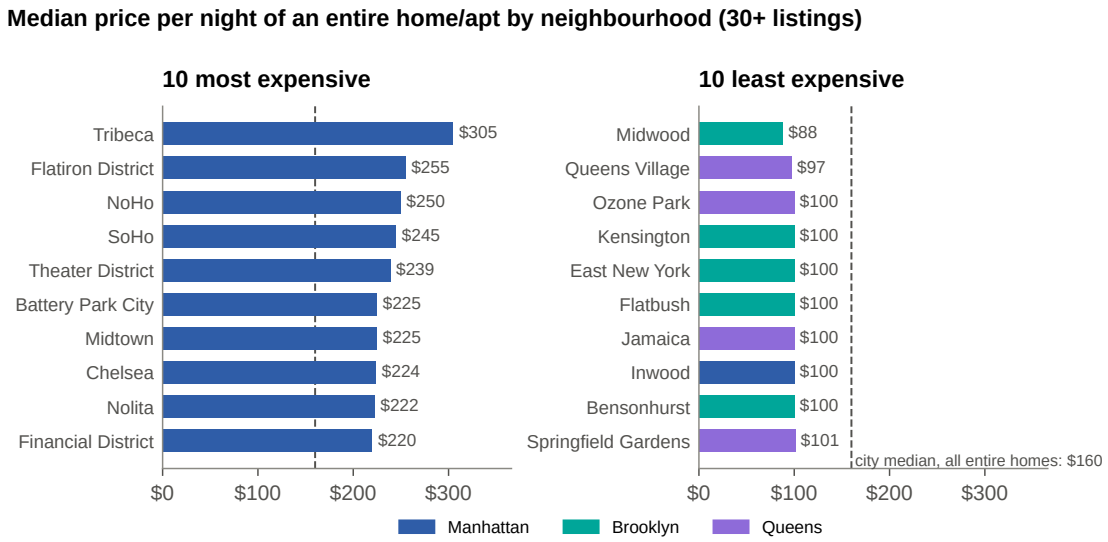


Figure 7: Median price of an entire home by neighbourhood (30+ listings): the 10 most and the 10 least expensive.

By neighbourhood, for an entire home, **Tribeca ($305)**, the Flatiron District, NoHo and SoHo are the most expensive; Midwood ($88) and Queens Village ($97) the least expensive: a ratio of **3.5 to 1** for the same type of product.

4.5 Is there a relationship between price, ratings and reviews? (Q4)

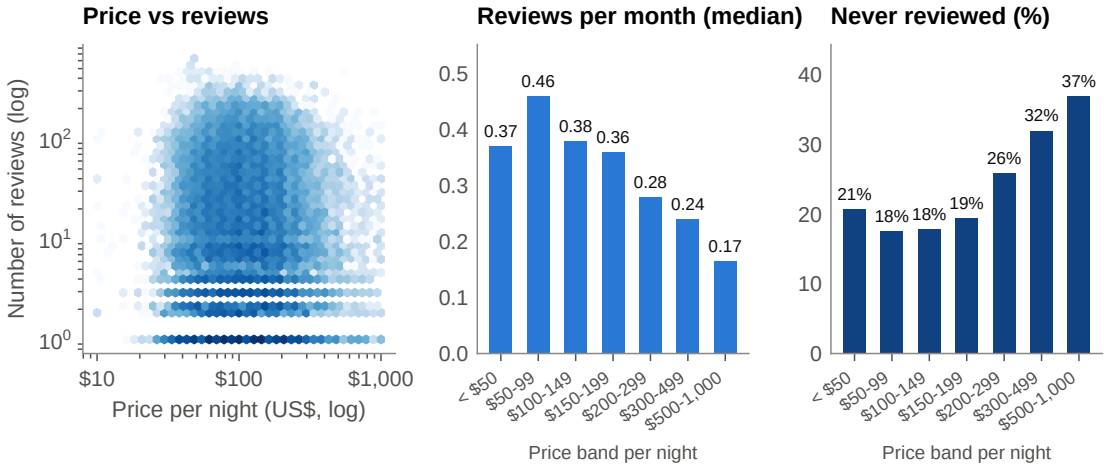


Figure 8: Price and number of reviews of every reviewed listing (left, hexbin, log scales), median review rate (middle) and share of never-reviewed listings (right) by price band.

The file has **no rating column**, so the question is answered with reviews. **Price and reviews are almost unrelated**: Spearman $\rho = -0.05$ with the number of reviews and with the review rate (significant because the sample is large, negligible in size); inside each room type ρ stays between -0.13 and 0.00. The \$50-99 band has the highest median review rate (0.46 per month); above \$200 the rate falls to 0.17 for \$500-1,000, and the share of **never-reviewed** listings rises

from 18% around \$50-150 to **37% above \$500**. Many reviews do not justify a higher price, and demand concentrates on affordable listings.

4.6 Which factors seem to affect price? (Q5)

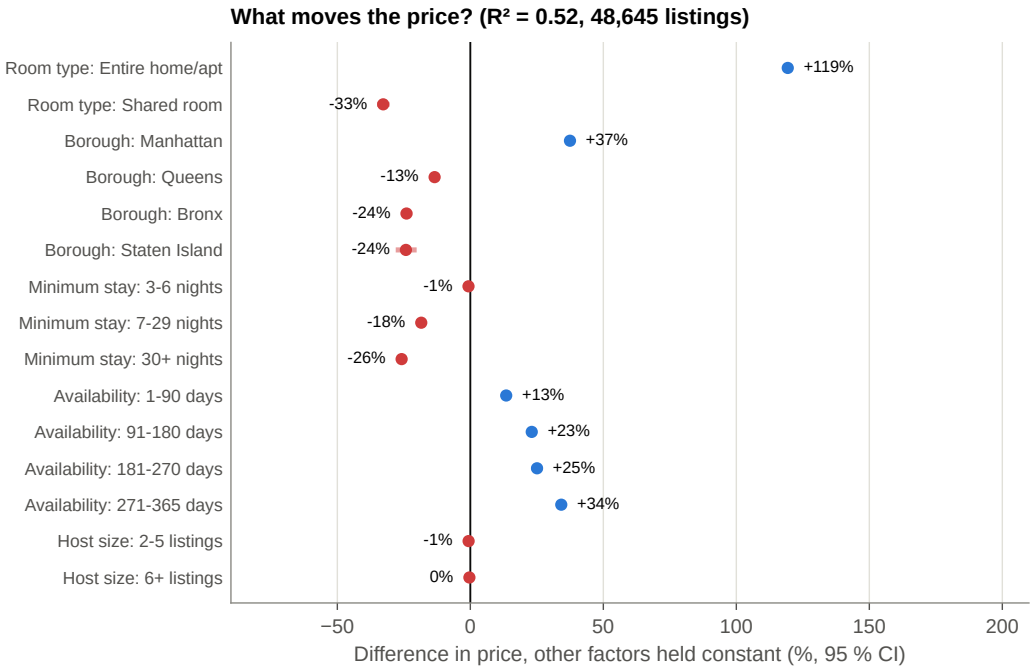


Figure 9: Effect of each factor on the price per night, other factors held constant (multiple regression of the log price, 95% confidence intervals). Reference: private room, Brooklyn, 1-2 nights, 0 days available, host with 1 listing.

A multiple regression of the log price measures each factor with the others held constant. It explains **52%** of the variation of the price (59% with the 221 neighbourhoods instead of the five boroughs).

- **Room type is the strongest factor**: alone it explains 41% of the variation. An entire home costs **+119%** compared with a private room, a shared room -33%.
- **Location comes second**: the neighbourhood alone explains 29%, the borough 14%. Compared with Brooklyn, a comparable listing costs **+37% in Manhattan**, -13% in Queens and -24% in the Bronx and Staten Island.
- **Minimum stay**: listings that require 30+ nights are 26% cheaper per night (7-29 nights: -18%): monthly rentals with a lower nightly rate.
- **Availability**: listings open most of the year are 13% to 34% more expensive than closed ones, the pattern of full-time listings.
- **Host size has no effect** once room type and location are taken into account, and more reviews go with slightly lower prices (-5% per unit of log reviews).

The remaining 48% depends on what the file does not contain: size, quality, amenities.

4.7 Which areas or listings perform best? (Q6)

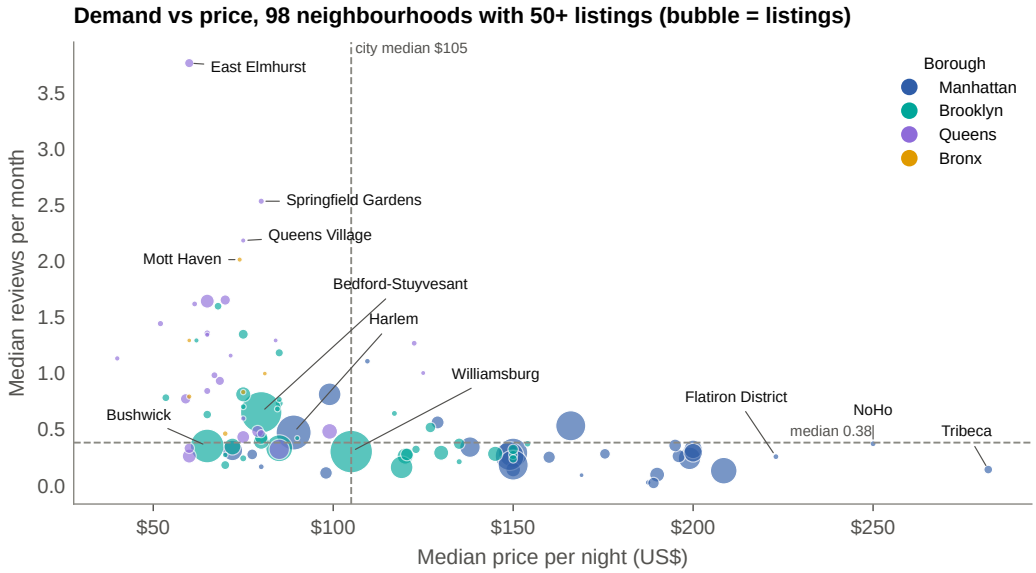


Figure 10: Median price and median review rate of the 98 neighbourhoods with 50+ listings (bubble = number of listings); dashed lines: city medians.

Performance is measured by demand, the review rate. Among the 98 neighbourhoods with 50+ listings, the highest median review rates are **East Elmhurst (3.76 reviews per month, next to LaGuardia airport)** and **Springfield Gardens (2.53, next to JFK)**, then Queens Village, Mott Haven (Bronx), Jamaica and Flushing, all at a median price of $60 to $80: **cheap and busy**. Large central areas (Williamsburg, Midtown, the Upper West Side) have many listings but low median rates (0.13 to 0.30), and the expensive Manhattan neighbourhoods (Tribeca, NoHo, Flatiron) sit in the lower right corner. At the listing level, 63 of the 100 listings with the highest review rate are in **Queens** and 64 are **private rooms**, often titled after the airport ("5 mins from JFK", "LGA airport"). The two Theater District rooms at the top (58.5 and 28 reviews per month) look like hotel rooms sold on Airbnb. Two models work: **volume** (affordable rooms near transport) and **value** (expensive central homes, fewer stays).

4.8 Are there noticeable patterns or trends? (Q7)

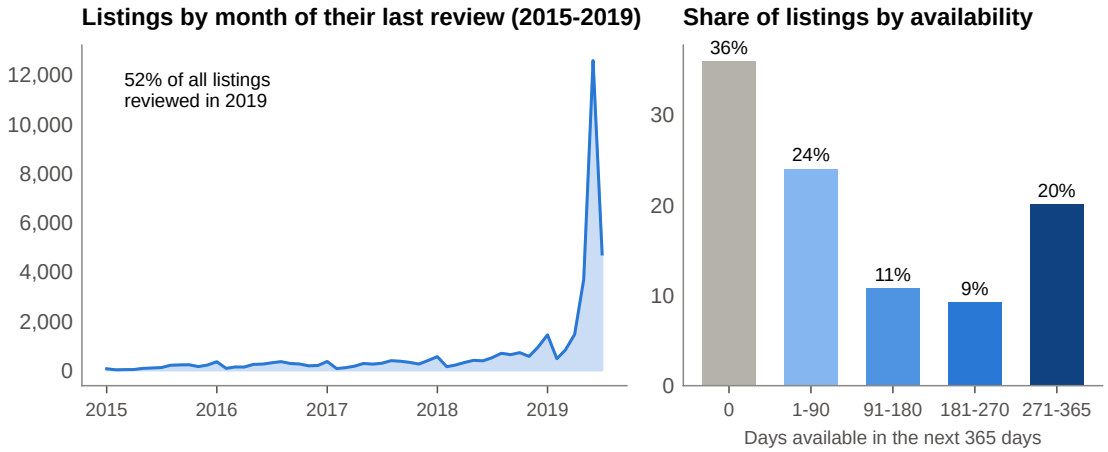


Figure 11: Listings by month of their last review (left) and share of listings by availability in the next 365 days (right).

A large part of the supply is inactive. Only **60%** of the listings were reviewed in the last 12 months and 20% were never reviewed; the last review of 52% of all listings falls in 2019, with a sharp peak in June-July 2019 (the active listings). **36% of the listings have 0 days of availability**, while 20% are open almost all year. Demand goes with availability and short stays: the review rate rises with availability ($\rho = +0.30$) and falls with the minimum stay ($\rho = -0.25$).

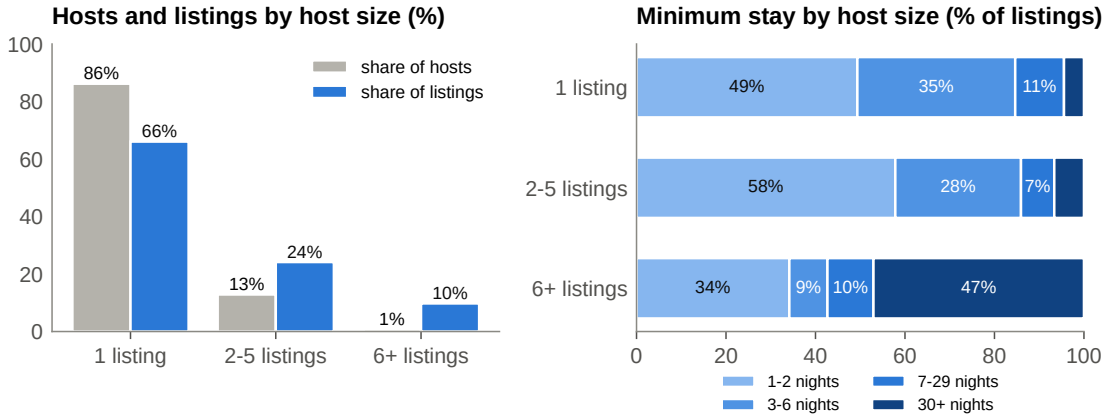


Figure 12: Share of hosts and of listings by host size (left) and minimum stay by host size (right).

There are two kinds of hosts. 86% of the hosts have a single listing and hold 66% of the offer, with a median availability of only 6 days: occasional home sharing. At the other end, **343 hosts (0.9%) run 6 listings or more** and hold **10% of the offer** (4,755 listings), open a median of **310 days** a year, and **47% of their listings require 30+ nights**: professional operators of furnished apartments (the 10 largest hosts alone run 1,268 listings). The file is one snapshot, so trends over time can only be read through the date of the last review.

4.9 Statistical tests

Prices and reviews are skewed, so the tests are rank-based. With 48,645 listings even weak effects are significant; the size of the effect is what matters.

Table 5: Kruskal-Wallis tests on the price, Spearman correlations (95% bootstrap intervals) and Mann-Whitney test.

Test	Effect	Reading
Price by room type (Kruskal-Wallis)	$\varepsilon^2 = 0.46$	large
Price by neighbourhood (221 groups)	$\varepsilon^2 = 0.30$	large
Price by borough	$\varepsilon^2 = 0.14$	moderate
Price by host size / by minimum stay	$\varepsilon^2 = 0.03$ / 0.02	small
Price – number of reviews	$\rho = -0.05$ [-0.06, -0.04]	negligible
Price – reviews per month	$\rho = -0.05$ [-0.06, -0.05]	negligible
Reviews per month – availability	$\rho = +0.30$ [0.29, 0.31]	moderate
Reviews per month – minimum nights	$\rho = -0.25$ [-0.26, -0.24]	moderate
Entire homes: multi-listing vs single hosts (Mann-Whitney)	medians \$175 vs \$155	multi-listing hosts ask more

All price differences are significant ($p < 0.001$). The regression of Section 4.6 uses robust (HC3) standard errors; all effects of room type, borough, minimum stay (except 3-6 nights) and availability are significant, host size is not. These are associations, not causes.

5 Important visualizations

Each question has one visual chosen for what it must show.

Table 6: Choice of the visuals.

Question	Visual	Why
Q1 listing types	bars + 100% stacked bars	counts with the median price on each bar; the mix in each borough
Q2 active areas	side-by-side bars, map	supply and demand next to each other; the map shows where
Q3 prices by area	histogram, box plots, ranked bars	skewed prices need a log scale and the spread, not only an average
Q4 price and reviews	hexbin + two bar charts	38,726 points would hide each other; bands show the trend, one scale per chart
Q5 price factors	dot plot with intervals	one effect per factor, other factors held constant
Q6 best performers	bubble chart	price, demand and size of each neighbourhood at once
Q7 patterns	time line, bars, stacked bars	recency, availability and host profiles

The **five most important visuals** are: the room-type mix (Figure 1), because it shows that the

market is two products; the supply and demand by borough (Figure 2), because Brooklyn leads demand; the price by borough and room type (Figure 6); the price factors (Figure 9), because it ranks what sets the price; and the demand-price bubble chart (Figure 10), because it shows where listings perform best. The figures follow one standard: one fixed colour per borough (a categorical palette checked for colour blindness, the same hues in the notebook, the dashboard and the presentation), a fixed colour per room type (entire home coral, private room cyan, shared room gold), labelled values placed so that no text overlaps another, one scale per chart, medians rather than means, and the base of each chart in its title.

6 Dashboard

The Power BI dashboard (`powerbi/NYC_Airbnb_Dashboard.pbix`) is built on the two cleaned tables of Section 3, the listings and the cleaning log, so every figure it shows is a figure of this report. Its model holds 44 DAX measures: the indicators of the analysis (listings, hosts, median and average price, reviews per listing, share of listings reviewed in the last 12 months, share of multi-listing hosts...), the medians restricted to groups large enough to be compared (30 listings for a price, 50 for a review rate), the city median used as a reference, and the counts of the cleaning log. Its seven pages follow the questions of the brief (Table 7).

Table 7: Pages of the dashboard.

Page	Content
1. Home	photo of the New York skyline; eight KPI cards (listings, hosts, median and average price, reviews, reviews per listing, share active in the last 12 months, neighbourhoods); word cloud of the neighbourhoods by number of listings; three picture tiles that open the analysis pages; a scrolling band of the largest neighbourhoods
2. Map & Boroughs	map of the 48,645 listings by borough; listings by borough; listing types; median price by borough (Q1, Q2)
3. Prices	median price by borough and room type; listings by price band; gauge of the median against the city median; the ten most expensive neighbourhoods; median price by minimum stay (Q3, Q5)
4. Demand	demand against price by neighbourhood; the ten busiest listings; median reviews per month by borough and by price band; listings by year of last review (Q2, Q4, Q6)
5. Hosts	shares of hosts and of listings by host size; minimum stay by host size; listings per host and hosts with 6 or more listings; days open in the year; share of multi-listing hosts by borough (Q7)
6. Insights	the nine findings and the recommendations
7. Data Pipeline	raw, removed and clean listings, share kept, cleaning steps; the steps from the supplied file to the model and the cleaning log

Interactivity. Five dropdown slicers (borough, neighbourhood, room type, price band, minimum stay) run along the top of the four analysis pages and keep their selection from one page

to the next. Clicking a bar, a slice or a point filters the other visuals of the page, and the cards, the gauge and the top-ten visuals follow every selection: the gauge, for example, compares the median of the selected listings with the median of the whole city. A page navigator in the header links the seven pages, and the picture tiles of the Home page open the Map, Prices and Demand pages. Tooltips add a second figure to each bar (for example the share of listings and the median price of a borough), and every visual carries an alternative text for screen readers.

Design. A dark navy theme with a single blue accent: a header band with a soft blue glow, the logo, the title and the page navigator; framed visuals with their title; KPI cards with rounded borders. Single measures are blue; a borough and a room type keep the same hue on every page as in the figures of this report, lightened for the dark background. The Home page uses the photo of the New York skyline supplied with the project, and the map plots each listing on the map of New York City supplied with the dataset. The icons are Microsoft's 3D Fluent Emoji (MIT licence). Cards, axes and tables use business labels ("Median price", "Share of listings") rather than the technical names of the measures; value axes are removed where the values are written on the bars. In the model, the 44 measures are grouped in display folders (key figures, prices, demand, hosts and availability, shares, data quality) with a description, and the technical sort columns are hidden.



Figure 13: Home: 48,645 listings, 37,283 hosts, a median of $105 and an average of $141.31 per night, 1,136,096 reviews (23.4 per listing), 59.8% of listings active in the last 12 months, 221 neighbourhoods; Williamsburg and Bedford-Stuyvesant are the largest neighbourhoods.

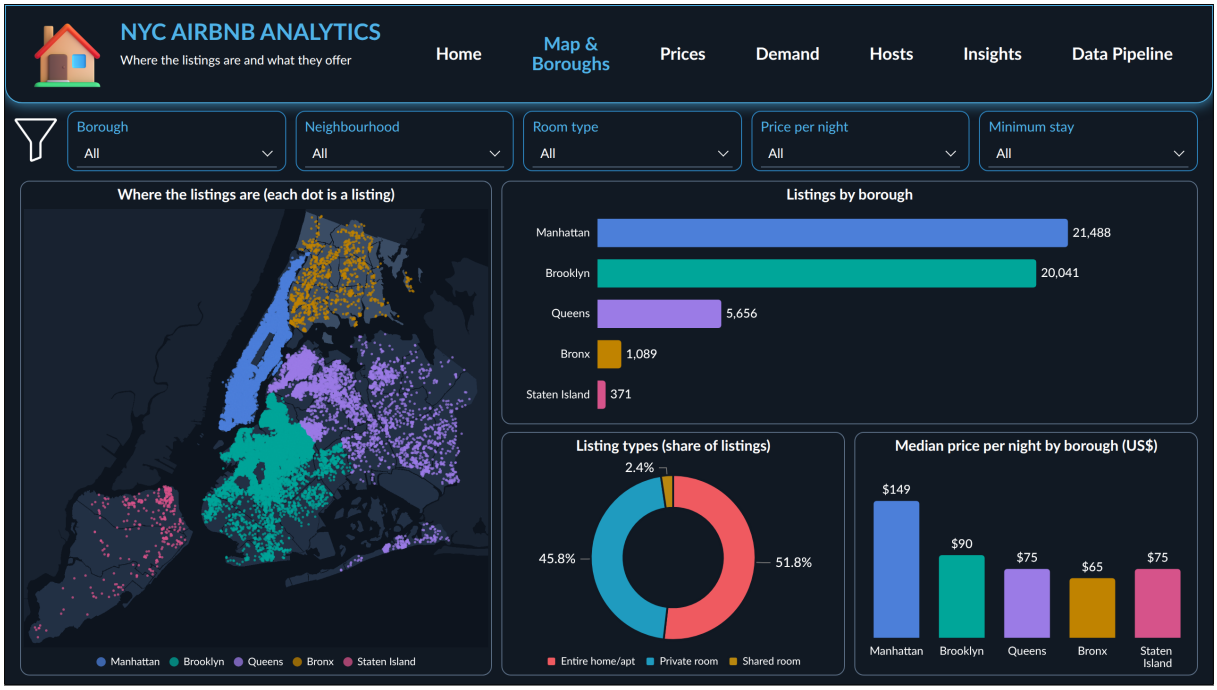


Figure 14: Map & Boroughs: Manhattan (21,488) and Brooklyn (20,041) hold 85% of the listings; 51.8% entire homes and 45.8% private rooms; median of $149 in Manhattan and $65 in the Bronx.



Figure 15: Prices: an entire home costs $190 in Manhattan and $100 in the Bronx; one listing in three asks $50 to $99; without a filter the gauge sits on the city median of $105; Tribeca is the most expensive neighbourhood ($282, all room types).



Figure 16: Demand: the cheap neighbourhoods of Queens get the most reviews a month; demand is highest in Staten Island (1.00) and lowest in Manhattan (0.29), and falls above $100 a night; 25,148 listings were last reviewed in 2019.



Figure 17: Hosts: 86.2% of hosts have one listing and hold 66.1% of the listings; the 343 hosts with 6 or more hold 9.8%, 47% of them with stays of 30 nights or more; 35.9% of calendars closed.

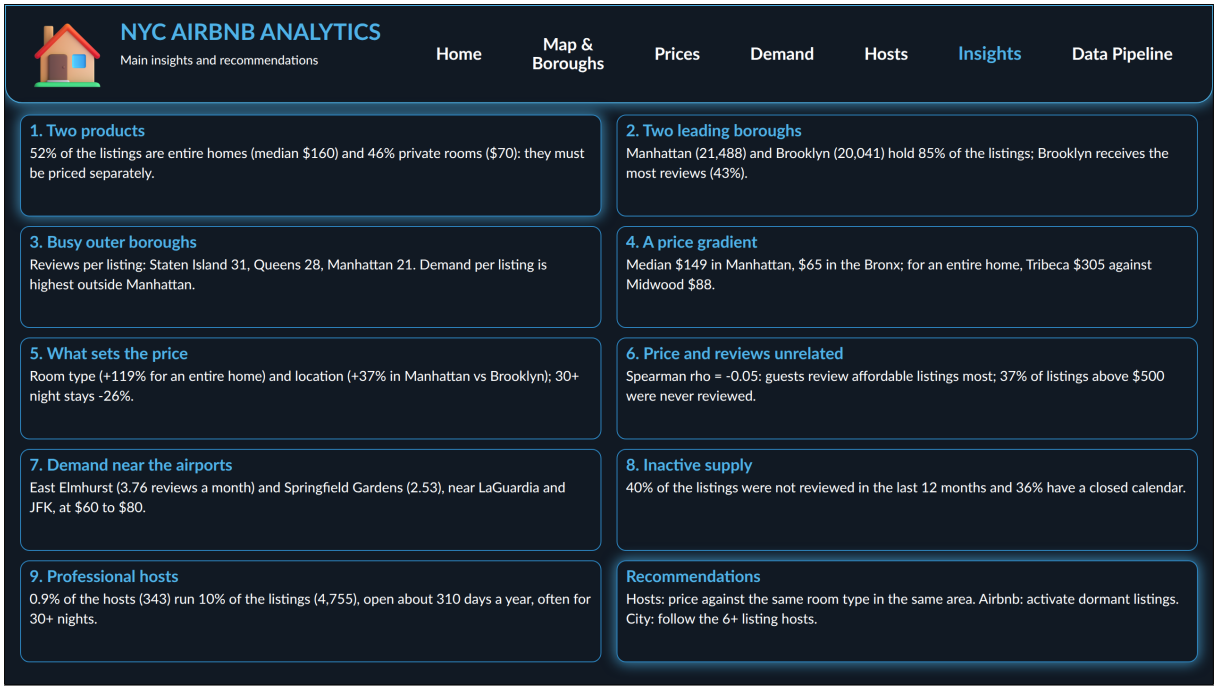


Figure 18: Insights: the nine findings and the recommendations for hosts, Airbnb and the city.

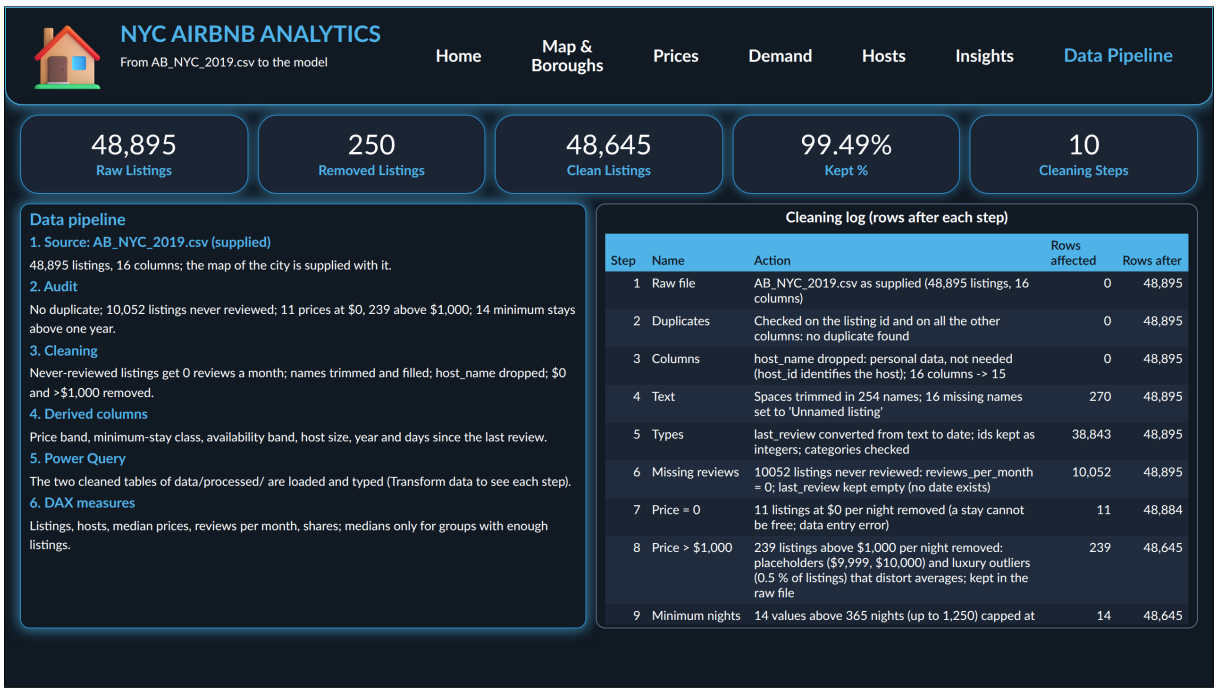


Figure 19: Data Pipeline: 48,895 raw listings, 250 removed, 48,645 kept (99.49%), ten cleaning steps.

7 Key findings

1. **Two products**: entire homes (52%, median $160) and private rooms (46%, $70).
2. **Two leading boroughs**: Manhattan and Brooklyn hold 85% of the listings; Brooklyn receives the most reviews (42.8%) and Bedford-Stuyvesant is the most reviewed neighbour-

hood.

3. **Intensive outer boroughs**: Staten Island 31 and Queens 28 reviews per listing, Manhattan 21.
4. **A price gradient**: median \$149 in Manhattan, \$65 in the Bronx; Tribeca \$305 and Midwood \$88 for an entire home.
5. **What sets the price**: room type (+119%), location (+37% in Manhattan vs Brooklyn), minimum stay (-26% for 30+ nights); host size has no effect.
6. **Price and reviews are unrelated** ($\rho = -0.05$); 37% of listings above \$500 were never reviewed.
7. **The busiest areas are near the airports**: East Elmhurst 3.76 and Springfield Gardens 2.53 reviews per month, at \$60-80.
8. **40% of the listings look inactive**; 36% have a closed calendar.
9. **A small professional segment**: 0.9% of hosts, 10% of listings, open 310 days a year, half with 30+ night stays.

8 Business insights

Table 8: Insights: what was discovered, what it means and why it is important.

What we discovered	What it means	Why it is important
Entire homes and private rooms make 98% of the offer, at \$160 and \$70	two products with two kinds of guests	prices and competitors must be compared within a room type
Brooklyn has fewer listings than Manhattan but more reviews	demand has moved beyond Manhattan	Brooklyn is as important as Manhattan for growth
Outer boroughs have more reviews per listing	supply is short where demand per listing is high	new hosts there face less competition
Price falls with the distance from Manhattan	location sets the price level	a city-wide average is a wrong benchmark
Room type and location explain most of the price; host size does not	hosts control the price through what and where they offer	pricing advice can be built on few variables
Price and reviews are unrelated; expensive listings are more often never reviewed	reviews follow affordability, not price	a high price does not bring more stays
Cheap rooms near the airports are the busiest listings	convenience and price win short stays	a clear, under-served niche
40% of the listings were not reviewed in 12 months	headline counts over-state the active supply	search results and market size are inflated
0.9% of hosts run 10% of the listings, open all year	a professional segment behaves differently	the city can target checks without burdening home sharers

9 Recommendations

For hosts

1. **Price against the right benchmark**: the median of the same room type in the same neighbourhood (Prices page), not the city average; a price far above it needs a visible reason.
2. **Consider the outer boroughs and transport hubs**: rooms near JFK and LaGuardia and in Queens are booked far more often, at a median of \$60-80.
3. **Open the calendar and shorten the minimum stay** if the goal is short stays: demand rises with availability ($\rho = +0.30$) and falls with the minimum stay ($\rho = -0.25$).

For Airbnb

4. **Activate dormant supply**: 20% of the listings were never reviewed and 36% have a closed calendar; reminders, pricing tips or delisting would make search results more reliable.

5. **Recruit hosts where demand per listing is high** (Queens, the Bronx, Staten Island) rather than in saturated central areas.
6. **Segment the offer by guest type**: affordable rooms for short, convenience-driven trips; entire homes in Manhattan for leisure and families; monthly rentals for longer stays.

For the city

7. **Monitor the professional segment**: host size, availability and minimum stay identify the operators of many entire homes (343 hosts with 4,755 listings) and allow targeted checks of the rules on short-term rentals.

For the next analysis

8. **Add ratings, bookings and property size** to move from demand proxies to satisfaction, occupancy and revenue.

Limitations

- No ratings: Q4 is answered with reviews, which measure demand, not satisfaction.
- No bookings or revenue: prices are asked prices, and not every guest leaves a review.
- No property size or quality, which is why the price model stops at $R^2 = 0.52$.
- One snapshot: time patterns come from the date of the last review only.
- Listings above $1,000 per night (0.5%) are excluded: the luxury segment is not covered.

10 Conclusion

From a raw file of 48,895 listings, this project produced a clean and documented dataset (48,645 listings, 10 logged steps), answered the seven questions of the brief with figures and tests, and delivered an interactive Power BI dashboard. The New York Airbnb market of 2019 is a market of **two products** (entire homes and private rooms), **concentrated in Manhattan and Brooklyn**, where **the room type and the location set the price**. Demand, measured by reviews, does not follow the price: it is strongest for **affordable rooms in the outer boroughs and near the airports**. A large part of the listed supply is inactive, and a small group of professional hosts runs one listing in ten. These findings give hosts a pricing benchmark, Airbnb a map of where supply is short or dormant, and the city a way to identify professional operators. Adding ratings, bookings and property size would turn this demand analysis into an analysis of satisfaction and revenue.

Reproducibility

Every number, table and figure of this report is produced by the notebook `NYC-Airbnb-Analytics.ipynb` from the files of `data/raw/`. The notebook writes the cleaned dataset

and the cleaning log (`data/processed/`), the result tables (`results/`) and the figures (`figures/`). The dashboard `powerbi/NYC_Airbnb_Dashboard.pbix` reads `data/processed/`. The final presentation is `presentation/NYC_Airbnb_Final_Presentation.pptx`. The project is published at `https://github.com/koffifidelek59-collab/nyc-airbnb-analytics`.

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